25901436 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCPHY201 - ELECTROMAGNETIC THEORY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **3** marks

1.	Calculate the work done in moving a charge of 1 C around the circumference of a circle of radius R ,starting and ending at the same point on the circle	[U] / [CO1]
2.	State Gauss's law in differential form and its relation to divergence.	[U] / [CO1]
3.	Can Gauss's law be applied to non-uniform charge distributions? Justify your answer briefly.	[U] / [CO1, CO2]
4.	What is electrostatic shielding?	[U] / [CO2]
5.	What is the curent in a line of charge?	[U] / [CO3]
6.	State the Coulob gauge condition for vector potentail in magnetostatics.	[U] / [CO3]
7.	Why are magnetic field lines always closed loops?	[U] / [CO3, CO4]
8.	What is the permittivity of free space?	[U] / [CO5]
9.	Define the term magnetization.suggest a method to enhance the magnetization in materials?	[K] / [CO5]

10. At what condition a neutral atom have an induced electric dipole moment [U] / [CO5]

11. Write Faraday's law in differential form and explain what it tells us about the curl of the electric field. [U] / [CO6]

12. What are free charge and bound charge in matter? [U] / [CO7]

[3x10 = 30]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Derive Poisson's equation and explain its physical significance. [U] / [CO1, CO2]

14. A charge of 6μC is placed at a point in an electric field of strength 3×10^4 N/C. Find the force experienced by the charge. [A] / [CO2]

15. Explain why the excess charge resides on the surface of a conductor? [U] / [CO2]

16. Differentiate between magnetic induction and magnetic flux. [U] / [CO3]

17. At what distance from a **20 A** current-carrying wire will the magnetic field be **4×10^{-5} T ?** [A] / [CO4]

18. Explain magnetic susceptibility and discuss its role in characterizing magnetic materials. [U] / [CO5]

19. The potential inside a uniformly polarized sphere is $V(r,\theta) = \frac{P}{3\epsilon_0} r\cos\theta$. What is the electric field inside the sphere. [E] / [CO5]

20. State and explain Gauss's law in dielectric media with mathematical form. [U] / [CO5]

21. Define electromotive force. Derive an expression for the terminal voltage of a cell in terms of emf, current, and internal resistance. [U] / [CO6]

22. Explain how a changing magnetic field produces an electric field [A] / [CO7]

[5x8 = 40]